

Linear Algebra Walkthrough

Lecture 2: CEF and Best Linear Predictor

Companion to Slides — PLSC 30700

This document expands every linear algebra step in Lecture 2 (“Linear Models: CEF and Best Linear Predictor”), connecting each derivation back to the corresponding slide. Where the slides use compact matrix notation, we unpack the matrices into their column-vector and element-wise forms.

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1 Bivariate BLP: Solving the First-Order Conditions

Slide Reference

“Solving for Best Linear Predictor (a and b)”

We want to find scalars a and b that minimize the mean squared prediction error:

$$M(a, b) = \mathbb{E}[(Y - (a + bX))^2].$$

1.1 Expanding the square

Write out the square explicitly:

$$\begin{aligned} M(a, b) &= \mathbb{E}[Y^2 - 2Y(a + bX) + (a + bX)^2] \\ &= \mathbb{E}[Y^2] - 2a\mathbb{E}[Y] - 2b\mathbb{E}[XY] + a^2 + 2ab\mathbb{E}[X] + b^2\mathbb{E}[X^2]. \end{aligned} \quad (1)$$

Every term uses the linearity of expectation: $\mathbb{E}[cZ] = c\mathbb{E}[Z]$ for a constant c .

1.2 FOC with respect to a

Differentiate (1) with respect to a , holding b fixed:

$$\frac{\partial M}{\partial a} = -2\mathbb{E}[Y] + 2a + 2b\mathbb{E}[X] = 0.$$

Solving:

$$\boxed{a = \mathbb{E}[Y] - b\mathbb{E}[X]}. \quad (2)$$

Interpretation. The intercept adjusts the prediction so that, on average, $a + b\mathbb{E}[X] = \mathbb{E}[Y]$ —the predictor passes through the point $(\mathbb{E}[X], \mathbb{E}[Y])$.

1.3 FOC with respect to b

Differentiate (1) with respect to b :

$$\frac{\partial M}{\partial b} = -2\mathbb{E}[XY] + 2a\mathbb{E}[X] + 2b\mathbb{E}[X^2] = 0.$$

Substitute $a = \mathbb{E}[Y] - b\mathbb{E}[X]$ from (2):

$$\begin{aligned} \mathbb{E}[XY] &= (\mathbb{E}[Y] - b\mathbb{E}[X])\mathbb{E}[X] + b\mathbb{E}[X^2] \\ &= \mathbb{E}[Y]\mathbb{E}[X] - b(\mathbb{E}[X])^2 + b\mathbb{E}[X^2] \\ &= \mathbb{E}[Y]\mathbb{E}[X] + b(\mathbb{E}[X^2] - (\mathbb{E}[X])^2). \end{aligned}$$

Rearranging:

$$\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = b(\mathbb{E}[X^2] - (\mathbb{E}[X])^2).$$

The left side is $\text{Cov}(X, Y)$ and the right side involves $\text{Var}(X)$:

$$\boxed{b = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}}. \quad (3)$$

1.4 The BLP in one formula

Substituting back:

$$\begin{aligned}\mathbb{E}^*[Y|X] &= a + bX = \mathbb{E}[Y] - b\mathbb{E}[X] + bX \\ &= \mathbb{E}[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - \mathbb{E}[X]).\end{aligned}$$

This is the population regression line, centered at $(\mathbb{E}[X], \mathbb{E}[Y])$ with slope $\text{Cov}(X, Y)/\text{Var}(X)$.

2 Variance of the BLP Error (Bivariate)

Slide Reference

“Deriving Variance of BLP at Minimized Values”

Define the projection error $e = Y - (\alpha + \beta X)$, where $\alpha = \mathbb{E}[Y] - \beta\mathbb{E}[X]$. We want $\text{Var}(e)$.

Step 1 (Rewrite the error). Since $\alpha = \mathbb{E}[Y] - \beta\mathbb{E}[X]$, subtracting $\alpha + \beta X$ from Y gives:

$$e = Y - \mathbb{E}[Y] + \beta\mathbb{E}[X] - \beta X = (Y - \mathbb{E}[Y]) - \beta(X - \mathbb{E}[X]).$$

So the error is a difference of two centered random variables.

Step 2 (Compute the variance). Using $\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2\text{Cov}(A, B)$ with $A = Y$ and $B = \beta X$:

$$\begin{aligned}\text{Var}(e) &= \text{Var}(Y - \beta X) \\ &= \text{Var}(Y) + \beta^2 \text{Var}(X) - 2\beta \text{Cov}(X, Y).\end{aligned}$$

The constant α drops out because $\text{Var}(Z + c) = \text{Var}(Z)$.

Step 3 (Substitute $\beta \text{Var}(X) = \text{Cov}(X, Y)$). From equation (3), $\beta = \text{Cov}(X, Y)/\text{Var}(X)$, so $\beta \text{Var}(X) = \text{Cov}(X, Y)$. Thus:

$$\begin{aligned}\text{Var}(e) &= \text{Var}(Y) + \beta^2 \text{Var}(X) - 2\beta \cdot \beta \text{Var}(X) \\ &= \text{Var}(Y) + \beta^2 \text{Var}(X) - 2\beta^2 \text{Var}(X) \\ &= \text{Var}(Y) - \beta^2 \text{Var}(X).\end{aligned}$$

$$\boxed{\sigma_e^2 = \sigma_Y^2 - \beta^2 \sigma_X^2.} \tag{4}$$

Interpretation. The error variance is the total variance of Y minus what the linear predictor explains. The quantity $\beta^2 \sigma_X^2 / \sigma_Y^2$ is the population R^2 .

3 Multivariate BLP: Expanding the MSPE

Slide Reference

“Deriving the Linear Projection Coefficient β ”

Now X is a $k \times 1$ random vector and β is a $k \times 1$ parameter vector. To make the algebra concrete, write:

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_{k-1} \\ 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{k-1} \\ \beta_k \end{pmatrix}.$$

The last element of X is the constant 1 (for the intercept), so β_k plays the role of the intercept.

The linear predictor is the inner product:

$$X'\beta = (X_1 \quad X_2 \quad \cdots \quad X_{k-1} \quad 1) \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_{k-1} \\ \beta_k \end{pmatrix} = \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_{k-1} X_{k-1} + \beta_k.$$

3.1 Step 1: Writing the MSPE

$$S(\beta) = \mathbb{E}[(Y - X'\beta)^2] = \mathbb{E}[(Y - X'\beta)(Y - X'\beta)]. \quad (5)$$

3.2 Step 2: Expanding the product

Note that $Y - X'\beta$ is a *scalar* random variable (1×1), so $(Y - X'\beta)^2 = (Y - X'\beta)(Y - X'\beta)$. Expand:

$$(Y - X'\beta)^2 = Y^2 - 2Y \cdot X'\beta + (X'\beta)^2.$$

We need to understand each term:

Term 1: Y^2 . Just the square of the scalar Y .

Term 2: $Y \cdot X'\beta$. Since $X'\beta$ is a scalar:

$$Y \cdot X'\beta = Y(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k) = \beta_1 Y X_1 + \beta_2 Y X_2 + \cdots + \beta_k Y.$$

Taking expectations and writing in matrix form:

$$\mathbb{E}[Y \cdot X'\beta] = \beta' \mathbb{E}[XY],$$

where $\mathbb{E}[XY]$ is the $k \times 1$ vector:

$$\mathbb{E}[XY] = \begin{pmatrix} \mathbb{E}[X_1 Y] \\ \mathbb{E}[X_2 Y] \\ \vdots \\ \mathbb{E}[Y] \end{pmatrix}.$$

(The last entry is $\mathbb{E}[1 \cdot Y] = \mathbb{E}[Y]$.)

Term 3: $(X'\beta)^2$. Since $X'\beta$ is a scalar, $(X'\beta)^2 = (X'\beta)(X'\beta) = (\beta'X)(X'\beta) = \beta'(XX')\beta$. The matrix XX' is $k \times k$. Expanding it column-by-column:

$$XX' = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ 1 \end{pmatrix} (X_1 \quad X_2 \quad \cdots \quad 1) = \begin{pmatrix} X_1^2 & X_1 X_2 & \cdots & X_1 \\ X_2 X_1 & X_2^2 & \cdots & X_2 \\ \vdots & \vdots & \ddots & \vdots \\ X_1 & X_2 & \cdots & 1 \end{pmatrix}.$$

Taking expectations element-by-element:

$$\mathbb{E}[XX'] = \mathbf{Q}_{XX} = \begin{pmatrix} \mathbb{E}[X_1^2] & \mathbb{E}[X_1 X_2] & \cdots & \mathbb{E}[X_1] \\ \mathbb{E}[X_2 X_1] & \mathbb{E}[X_2^2] & \cdots & \mathbb{E}[X_2] \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[X_1] & \mathbb{E}[X_2] & \cdots & 1 \end{pmatrix}.$$

Key fact: $\mathbf{Q}_{XX}$ is symmetric ($\mathbb{E}[X_i X_j] = \mathbb{E}[X_j X_i]$) and, by the regularity condition, positive definite.

3.3 Step 3: The MSPE in matrix form

Putting the three terms together under the expectation:

$$S(\beta) = \mathbb{E}[Y^2] - 2\beta'\mathbb{E}[XY] + \beta'\mathbb{E}[XX']\beta. \quad (6)$$

Remark 1. Notice the parallel with the bivariate case in Section 1. The scalar $\mathbb{E}[X^2]$ has become the matrix $\mathbb{E}[XX']$, and the scalar $\mathbb{E}[XY]$ has become the vector $\mathbb{E}[XY]$.

3.4 Step 4: Differentiating with respect to β

We use the matrix calculus rules:

- $\frac{\partial}{\partial \beta}(\beta' \mathbf{a}) = \mathbf{a}$ for constant vector $\mathbf{a}$.
- $\frac{\partial}{\partial \beta}(\beta' \mathbf{A} \beta) = 2\mathbf{A}\beta$ for symmetric matrix $\mathbf{A}$.

Applying these to (6):

$$\frac{\partial S}{\partial \beta} = -2\mathbb{E}[XY] + 2\mathbb{E}[XX']\beta.$$

Element by element. The j -th component of the gradient is:

$$\frac{\partial S}{\partial \beta_j} = -2\mathbb{E}[X_j Y] + 2 \sum_{i=1}^k \mathbb{E}[X_j X_i] \beta_i, \quad j = 1, \dots, k.$$

This is a system of k equations in k unknowns.

3.5 Step 5: Setting the FOC to zero and solving

$$\begin{aligned} 0 &= -2\mathbb{E}[XY] + 2\mathbb{E}[XX']\beta \\ \mathbb{E}[XY] &= \mathbb{E}[XX']\beta \\ \mathbf{Q}_{XY} &= \mathbf{Q}_{XX}\beta. \end{aligned} \tag{7}$$

Since $\mathbf{Q}_{XX}$ is positive definite (regularity condition 3), it is invertible. Premultiply both sides by $\mathbf{Q}_{XX}^{-1}$:

$$\boxed{\beta = \mathbf{Q}_{XX}^{-1} \mathbf{Q}_{XY} = \mathbb{E}[XX']^{-1} \mathbb{E}[XY].} \tag{8}$$

Written out for $k = 3$ (two regressors plus intercept):

$$\begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} \mathbb{E}[X_1^2] & \mathbb{E}[X_1 X_2] & \mathbb{E}[X_1] \\ \mathbb{E}[X_1 X_2] & \mathbb{E}[X_2^2] & \mathbb{E}[X_2] \\ \mathbb{E}[X_1] & \mathbb{E}[X_2] & 1 \end{pmatrix}^{-1} \begin{pmatrix} \mathbb{E}[X_1 Y] \\ \mathbb{E}[X_2 Y] \\ \mathbb{E}[Y] \end{pmatrix}.$$

4 The Projection Error is Uncorrelated with X

Slide Reference

“Linear Projection and Projection Error”

Define $e = Y - X'\beta$, so $Y = X'\beta + e$. We show $\mathbb{E}[Xe] = \mathbf{0}$.

4.1 The algebra

$$\begin{aligned} \mathbb{E}[Xe] &= \mathbb{E}[X(Y - X'\beta)] \\ &= \mathbb{E}[XY] - \mathbb{E}[XX']\beta. \end{aligned}$$

Why $\mathbb{E}[X \cdot X'\beta] = \mathbb{E}[XX']\beta$: The product $X \cdot X'\beta$ has j -th element:

$$X_j \cdot (X'\beta) = X_j \sum_{i=1}^k X_i \beta_i = \sum_{i=1}^k X_j X_i \beta_i.$$

Taking expectations: $\mathbb{E}[X_j \cdot X' \beta] = \sum_i \mathbb{E}[X_j X_i] \beta_i$, which is the j -th row of $\mathbb{E}[XX']\beta$.
 Now substitute $\beta = \mathbb{E}[XX']^{-1} \mathbb{E}[XY]$:

$$\begin{aligned} \mathbb{E}[Xe] &= \mathbb{E}[XY] - \mathbb{E}[XX'] \cdot \mathbb{E}[XX']^{-1} \mathbb{E}[XY] \\ &= \mathbb{E}[XY] - \mathbf{I}_k \cdot \mathbb{E}[XY] \\ &= \mathbb{E}[XY] - \mathbb{E}[XY] = \mathbf{0}. \end{aligned}$$

4.2 What these k equations say

Written as a vector equation:

$$\mathbb{E}[Xe] = \begin{pmatrix} \mathbb{E}[X_1 e] \\ \mathbb{E}[X_2 e] \\ \vdots \\ \mathbb{E}[1 \cdot e] \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

- **Equations 1 through $k-1$:** $\mathbb{E}[X_j e] = 0$ for each regressor. The error is uncorrelated with every regressor.
- **Equation k (the constant):** $\mathbb{E}[1 \cdot e] = \mathbb{E}[e] = 0$. The error has mean zero.

This is why including a constant matters: it forces $\mathbb{E}[e] = 0$.

5 Why $\mathbf{Q}_{XX}$ Must Be Positive Definite

Slide Reference

“The Design Matrix”

The BLP requires inverting $\mathbf{Q}_{XX} = \mathbb{E}[XX']$. We need this matrix to be positive definite.

5.1 Positive semi-definiteness is automatic

For any non-zero vector $\alpha \in \mathbb{R}^k$:

$$\alpha' \mathbf{Q}_{XX} \alpha = \alpha' \mathbb{E}[XX'] \alpha = \mathbb{E}[\alpha' XX' \alpha] = \mathbb{E}[(\alpha' X)^2] \geq 0.$$

Expanding $\alpha' X$:

$$\alpha' X = \alpha_1 X_1 + \alpha_2 X_2 + \cdots + \alpha_k \cdot 1.$$

This is a linear combination of the regressors. Its square is non-negative, so its expectation is non-negative.

5.2 Strict positive definiteness = no perfect collinearity

$\mathbf{Q}_{XX}$ is positive definite (not just semi-definite) if and only if $\mathbb{E}[(\alpha'X)^2] > 0$ for all $\alpha \neq \mathbf{0}$.

If $\mathbb{E}[(\alpha'X)^2] = 0$ for some $\alpha \neq \mathbf{0}$, then $\alpha'X = 0$ almost surely, meaning:

$$\alpha_1 X_1 + \alpha_2 X_2 + \cdots + \alpha_k X_k = 0 \quad \text{with probability 1.}$$

This is **perfect collinearity**: one regressor is an exact linear function of the others. In that case $\mathbf{Q}_{XX}$ is singular, β is not uniquely defined, and the BLP does not exist.

6 Multivariate Error Variance

Slide Reference

“Linear Predictor Error Variance”

Define $\sigma^2 = \mathbb{E}[e^2]$ where $e = Y - X'\beta$. We expand this using the matrices $\mathbf{Q}_{YY} = \mathbb{E}[Y^2]$, $\mathbf{Q}_{YX} = \mathbb{E}[YX']$ (a $1 \times k$ row vector), and $\mathbf{Q}_{XY} = \mathbb{E}[XY]$ (a $k \times 1$ column vector).

6.1 Expanding $\mathbb{E}[e^2]$

$$\begin{aligned} \sigma^2 &= \mathbb{E}[(Y - X'\beta)^2] \\ &= \mathbb{E}[Y^2] - 2\mathbb{E}[YX']\beta + \beta'\mathbb{E}[XX']\beta \\ &= \mathbf{Q}_{YY} - 2\mathbf{Q}_{YX}\beta + \beta'\mathbf{Q}_{XX}\beta. \end{aligned}$$

6.2 Substituting $\beta = \mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY}$

The second term:

$$2\mathbf{Q}_{YX}\beta = 2\mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY}.$$

(This is a $1 \times k$ times $k \times k$ times $k \times 1 = \text{scalar}$.)

The third term:

$$\begin{aligned} \beta'\mathbf{Q}_{XX}\beta &= (\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY})'\mathbf{Q}_{XX}(\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY}) \\ &= \mathbf{Q}_{YX}(\mathbf{Q}_{XX}^{-1})'\mathbf{Q}_{XX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY}. \end{aligned}$$

Since $\mathbf{Q}_{XX}$ is symmetric, $(\mathbf{Q}_{XX}^{-1})' = \mathbf{Q}_{XX}^{-1}$, so:

$$\beta'\mathbf{Q}_{XX}\beta = \mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\underbrace{\mathbf{Q}_{XX}\mathbf{Q}_{XX}^{-1}}_{\mathbf{I}_k}\mathbf{Q}_{XY} = \mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY}.$$

Combining:

$$\begin{aligned}\sigma^2 &= Q_{YY} - 2\mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY} + \mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY} \\ &= Q_{YY} - \mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY}.\end{aligned}$$

$$\boxed{\sigma^2 \equiv Q_{YY.X} = Q_{YY} - \mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY}.} \quad (9)$$

The notation $Q_{YY.X}$ is read “the variance of Y after projecting out X .”

6.3 Expanded for $k = 2$ (one regressor plus intercept)

Let $X = (X_1, 1)'$. Then:

$$\mathbf{Q}_{YX}\mathbf{Q}_{XX}^{-1}\mathbf{Q}_{XY} = \begin{pmatrix} \mathbb{E}[X_1Y] & \mathbb{E}[Y] \end{pmatrix} \begin{pmatrix} \mathbb{E}[X_1^2] & \mathbb{E}[X_1] \\ \mathbb{E}[X_1] & 1 \end{pmatrix}^{-1} \begin{pmatrix} \mathbb{E}[X_1Y] \\ \mathbb{E}[Y] \end{pmatrix}.$$

This is the multivariate generalization of $\beta^2 \text{Var}(X)$ from Section 2.

7 Intercepts, Slopes, and Demeaning

Slide Reference

“Intercepts and Slopes”

The slides separate the constant from X and write $Y = X'\beta + \alpha + e$, then show that demeaning produces $\beta = \text{Var}[X]^{-1} \text{Cov}(X, Y)$.

7.1 Setup: separating the intercept

Let $X = (X_1, \dots, X_{k-1})'$ (no constant) and write:

$$Y = X'\beta + \alpha + e.$$

From the FOC for α (analogous to Section 1):

$$\alpha = \mathbb{E}[Y] - \mathbb{E}[X]'\beta = \mu_Y - \mu_X'\beta.$$

7.2 Demeaning

Substitute α :

$$\begin{aligned}Y &= X'\beta + (\mu_Y - \mu_X'\beta) + e \\ Y - \mu_Y &= (X - \mu_X)'\beta + e.\end{aligned}$$

Now define the centered variables $\tilde{Y} = Y - \mu_Y$ and $\tilde{X} = X - \mu_X$:

$$\tilde{Y} = \tilde{X}'\beta + e.$$

The FOC for the demeaned problem gives:

$$\begin{aligned}\mathbb{E}[\tilde{X}\tilde{Y}] &= \mathbb{E}[\tilde{X}\tilde{X}']\beta \\ \text{Cov}(X, Y) &= \text{Var}(X) \cdot \beta.\end{aligned}$$

Here:

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)(X - \mu_X)'] = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

and

$$\text{Cov}(X, Y) = \begin{pmatrix} \text{Cov}(X_1, Y) \\ \text{Cov}(X_2, Y) \\ \vdots \end{pmatrix}.$$

Solving:

$$\boxed{\beta = \text{Var}(X)^{-1} \text{Cov}(X, Y).} \quad (10)$$

Connection to the second-moment formula. The full formula $\beta = \mathbb{E}[XX']^{-1}\mathbb{E}[XY]$ with the constant included is *algebraically equivalent* to computing the slope via centered moments and then recovering the intercept. Demeaning “absorbs” the intercept.

8 Partitioned Regression

Slide Reference

“Partitions and Regression Sub-Vectors,” “Partitioned Matrix Inversion and Regression,” “Interpreting Multivariate Regression,” “Demonstration: Iterated Projection”

8.1 Setup

Partition X into two sub-vectors:

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

where X_1 is $k_1 \times 1$ and X_2 is $k_2 \times 1$ (with $k_1 + k_2 = k$).

For concreteness, suppose $k_1 = 1$ (a single variable of interest) and $k_2 = 2$ (two controls plus the constant):

$$X_1 = (X_1), \quad X_2 = \begin{pmatrix} X_2 \\ X_3 \\ 1 \end{pmatrix}.$$

The linear projection is:

$$Y = X_1'\beta_1 + X_2'\beta_2 + e, \quad \mathbb{E}[Xe] = 0.$$

8.2 Partitioning $\mathbf{Q}_{XX}$ and $\mathbf{Q}_{XY}$

The normal equation $\mathbf{Q}_{XY} = \mathbf{Q}_{XX}\beta$ becomes:

$$\begin{pmatrix} \mathbf{Q}_{1Y} \\ \mathbf{Q}_{2Y} \end{pmatrix} = \begin{pmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

where:

$$\begin{aligned} \mathbf{Q}_{11} &= \mathbb{E}[X_1 X_1'] & (k_1 \times k_1), & & \mathbf{Q}_{12} &= \mathbb{E}[X_1 X_2'] & (k_1 \times k_2), \\ \mathbf{Q}_{21} &= \mathbb{E}[X_2 X_1'] & (k_2 \times k_1), & & \mathbf{Q}_{22} &= \mathbb{E}[X_2 X_2'] & (k_2 \times k_2). \end{aligned}$$

This is two equations:

$$\mathbf{Q}_{1Y} = \mathbf{Q}_{11}\beta_1 + \mathbf{Q}_{12}\beta_2, \tag{11}$$

$$\mathbf{Q}_{2Y} = \mathbf{Q}_{21}\beta_1 + \mathbf{Q}_{22}\beta_2. \tag{12}$$

8.3 Solving for β_1 via elimination

Step 4 (Solve (12) for β_2).

$$\beta_2 = \mathbf{Q}_{22}^{-1}(\mathbf{Q}_{2Y} - \mathbf{Q}_{21}\beta_1).$$

Step 5 (Substitute into (11)).

$$\begin{aligned} \mathbf{Q}_{1Y} &= \mathbf{Q}_{11}\beta_1 + \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}(\mathbf{Q}_{2Y} - \mathbf{Q}_{21}\beta_1) \\ \mathbf{Q}_{1Y} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y} &= (\mathbf{Q}_{11} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21})\beta_1. \end{aligned}$$

Step 6 (Identify the “residualized” quantities). Define:

$$\begin{aligned} \mathbf{Q}_{11 \cdot 2} &\equiv \mathbf{Q}_{11} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}, \\ \mathbf{Q}_{1Y \cdot 2} &\equiv \mathbf{Q}_{1Y} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y}. \end{aligned}$$

Then:

$$\boxed{\beta_1 = \mathbf{Q}_{11 \cdot 2}^{-1} \mathbf{Q}_{1Y \cdot 2}}. \tag{13}$$

By symmetry: $\beta_2 = \mathbf{Q}_{22 \cdot 1}^{-1} \mathbf{Q}_{2Y \cdot 1}$.

8.4 What $\mathbf{Q}_{11 \cdot 2}$ and $\mathbf{Q}_{1Y \cdot 2}$ mean: the iterated projection

Slide Reference

“Demonstration: Iterated Projection”

Step A: Regress X_1 on X_2 .

$$X_1 = X_2'\gamma_2 + u_1, \quad \mathbb{E}[X_2 u_1] = 0.$$

The coefficient is $\gamma_2 = \mathbf{Q}_{22}^{-1}\mathbf{Q}_{21}$ and the residual $u_1 = X_1 - X_2'\gamma_2$ is the part of X_1 that X_2 cannot predict.

Step B: Compute $\mathbb{E}[u_1^2]$.

$$\begin{aligned}
 \mathbb{E}[u_1^2] &= \mathbb{E}[(X_1 - X_2'\gamma_2)^2] \\
 &= \mathbb{E}[X_1^2] - 2\gamma_2'\mathbb{E}[X_2X_1] + \gamma_2'\mathbb{E}[X_2X_2']\gamma_2 \\
 &= \mathbf{Q}_{11} - 2\mathbf{Q}_{21}'\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21} + \mathbf{Q}_{21}'\mathbf{Q}_{22}^{-1}\mathbf{Q}_{22}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21} \\
 &= \mathbf{Q}_{11} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{21} = \mathbf{Q}_{11\cdot 2}.
 \end{aligned}$$

So $\mathbf{Q}_{11\cdot 2}$ is the variance of X_1 after removing what X_2 predicts.

Step C: Compute $\mathbb{E}[u_1Y]$.

$$\begin{aligned}
 \mathbb{E}[u_1Y] &= \mathbb{E}[(X_1 - X_2'\gamma_2)Y] \\
 &= \mathbb{E}[X_1Y] - \gamma_2'\mathbb{E}[X_2Y] \\
 &= \mathbf{Q}_{1Y} - \mathbf{Q}_{12}\mathbf{Q}_{22}^{-1}\mathbf{Q}_{2Y} = \mathbf{Q}_{1Y\cdot 2}.
 \end{aligned}$$

So $\mathbf{Q}_{1Y\cdot 2}$ is the covariance of the residualized X_1 with Y .

Conclusion:

$$\beta_1 = \mathbf{Q}_{11\cdot 2}^{-1}\mathbf{Q}_{1Y\cdot 2} = \frac{\mathbb{E}[u_1Y]}{\mathbb{E}[u_1^2]}. \quad (14)$$

When X_1 is a single variable (scalar), this is just $\text{Cov}(u_1, Y) / \text{Var}(u_1)$: the bivariate slope of Y on the part of X_1 that is unique—not explained by X_2 .

This is the Frisch–Waugh–Lovell intuition: The coefficient on X_1 in a multivariate regression is identical to the coefficient from a simple regression of Y on the residuals from regressing X_1 on all other regressors.

9 Omitted Variable Bias

Slide Reference

“Omitted Variable Bias”

9.1 Setup

Long regression (the correctly specified model):

$$Y = X_1'\beta_1 + X_2'\beta_2 + e, \quad \mathbb{E}[Xe] = 0.$$

Short regression (omitting X_2):

$$Y = X_1'\gamma_1 + u, \quad \mathbb{E}[X_1u] = 0.$$

The short-regression coefficient is:

$$\gamma_1 = (\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1Y].$$

9.2 Deriving the bias

Substitute $Y = X_1'\beta_1 + X_2'\beta_2 + e$ into the formula for γ_1 :

$$\begin{aligned}\gamma_1 &= (\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1(X_1'\beta_1 + X_2'\beta_2 + e)] \\ &= (\mathbb{E}[X_1X_1'])^{-1}[\mathbb{E}[X_1X_1']\beta_1 + \mathbb{E}[X_1X_2']\beta_2 + \mathbb{E}[X_1e]].\end{aligned}$$

Term by term:

First term: $(\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1X_1']\beta_1 = \mathbf{I}_{k_1}\beta_1 = \beta_1$.

Second term: $(\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1X_2'] = \Gamma_{12}$, a $k_1 \times k_2$ matrix.

What is Γ_{12} ? It is the matrix of coefficients from regressing each variable in X_2 on X_1 . The (i, j) entry tells you how much the j -th element of X_2 is associated with the i -th element of X_1 .

Third term: $(\mathbb{E}[X_1X_1'])^{-1}\mathbb{E}[X_1e] = (\mathbb{E}[X_1X_1'])^{-1} \cdot \mathbf{0} = \mathbf{0}$, because $\mathbb{E}[X_1e] = \mathbf{0}$ from the long regression (the first k_1 rows of $\mathbb{E}[Xe] = \mathbf{0}$).

Result:

$$\boxed{\gamma_1 = \beta_1 + \Gamma_{12}\beta_2.} \tag{15}$$

The **omitted variable bias** is $\Gamma_{12}\beta_2$.

9.3 Expanded for the scalar case

If X_1 and X_2 are both scalars:

$$\Gamma_{12} = \frac{\mathbb{E}[X_1X_2]}{\mathbb{E}[X_1^2]} = \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)} \quad (\text{when demeaned}),$$

and the bias is:

$$\text{Bias} = \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)} \cdot \beta_2.$$

- If X_1 and X_2 are positively correlated ($\Gamma_{12} > 0$) and X_2 has a positive effect on Y ($\beta_2 > 0$), then $\gamma_1 > \beta_1$: the short regression **overstates** the effect.
- If the correlation and the effect have opposite signs, the bias goes the other direction.
- If X_1 and X_2 are uncorrelated ($\Gamma_{12} = 0$), there is no bias: $\gamma_1 = \beta_1$.

10 CEF Algebra: Variance Decomposition and Theorem Proofs

Slide Reference

“Covariance of CEF and Its Error,” “Theorem 2.7: CEF as Best Predictor”

The CEF derivations are primarily probability rather than linear algebra, but they use algebraic expansions that parallel the BLP derivations.

10.1 Covariance of $m(X)$ and e

We show $\text{Cov}(e, m(X)) = 0$ where $e = Y - m(X)$ and $m(X) = \mathbb{E}[Y|X]$.

Step 7 (Compute $\mathbb{E}[e \cdot m(X)]$). *Use the law of iterated expectations, conditioning on X :*

Step 8 (Compute the covariance).

$$\begin{aligned}\text{Cov}(e, m(X)) &= \mathbb{E}[e \cdot m(X)] - \mathbb{E}[e]\mathbb{E}[m(X)] \\ &= 0 - 0 \cdot \mathbb{E}[m(X)] = 0.\end{aligned}$$

(We used $\mathbb{E}[e] = 0$ from the LIE.)

Step 9 (Variance decomposition). Since $Y = m(X) + e$ and $\text{Cov}(m(X), e) = 0$:

$$\text{Var}(Y) = \text{Var}(m(X)) + \text{Var}(e) + 2\text{Cov}(m(X), e) = \text{Var}(m(X)) + \text{Var}(e).$$

10.2 Theorem 2.7: CEF is the best predictor

We show $\mathbb{E}[(Y - g(X))^2] \geq \mathbb{E}[(Y - m(X))^2]$ for any function g .

Step 10 (Decompose using $Y = m(X) + e$).

$$Y - g(X) = e + (m(X) - g(X)).$$

Step 11 (Expand the square).

$$\begin{aligned}\mathbb{E}[(Y - g(X))^2] &= \mathbb{E}[(e + (m(X) - g(X)))^2] \\ &= \mathbb{E}[e^2] + 2\mathbb{E}[e(m(X) - g(X))] + \mathbb{E}[(m(X) - g(X))^2].\end{aligned}$$

Step 12 (The cross term vanishes). Since $m(X) - g(X)$ is a function of X , Theorem 2.4.4 gives $\mathbb{E}[e \cdot h(X)] = 0$ for any measurable $h(X)$. So $\mathbb{E}[e(m(X) - g(X))] = 0$.

Step 13 (Conclude).

$$\mathbb{E}[(Y - g(X))^2] = \underbrace{\mathbb{E}[e^2]}_{\mathbb{E}[(Y - m(X))^2]} + \underbrace{\mathbb{E}[(m(X) - g(X))^2]}_{\geq 0} \geq \mathbb{E}[(Y - m(X))^2].$$

Equality holds if and only if $g(X) = m(X)$ almost surely.

11 Partitioned Matrix Inverse Formula

Slide Reference

“Partitioned Matrix Inversion and Regression”

The slide states the block inverse of $\mathbf{Q}_{XX}$:

$$\mathbf{Q}_{XX}^{-1} = \begin{pmatrix} \mathbf{Q}_{11.2}^{-1} & -\mathbf{Q}_{11.2}^{-1} \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} \\ -\mathbf{Q}_{22.1}^{-1} \mathbf{Q}_{21} \mathbf{Q}_{11}^{-1} & \mathbf{Q}_{22.1}^{-1} \end{pmatrix},$$

where $\mathbf{Q}_{11.2} = \mathbf{Q}_{11} - \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} \mathbf{Q}_{21}$ and $\mathbf{Q}_{22.1} = \mathbf{Q}_{22} - \mathbf{Q}_{21} \mathbf{Q}_{11}^{-1} \mathbf{Q}_{12}$.

11.1 Verification

We verify by checking $\mathbf{Q}_{XX} \cdot \mathbf{Q}_{XX}^{-1} = \mathbf{I}$. Denote the proposed inverse as $\mathbf{M}$:

$$\begin{pmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_{k_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{k_2} \end{pmatrix}.$$

Upper-left block:

$$\mathbf{Q}_{11} \mathbf{Q}_{11.2}^{-1} + \mathbf{Q}_{12} (-\mathbf{Q}_{22.1}^{-1} \mathbf{Q}_{21} \mathbf{Q}_{11}^{-1}).$$

This requires extended algebra involving the Schur complement identity. The key fact is:

$$\mathbf{Q}_{11.2}^{-1} = (\mathbf{Q}_{11} - \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} \mathbf{Q}_{21})^{-1},$$

which is the Schur complement of $\mathbf{Q}_{22}$ in the block matrix. The formula is a standard result from block matrix algebra.

Upper-right block (checking it equals 0):

$$\begin{aligned} & \mathbf{Q}_{11} (-\mathbf{Q}_{11.2}^{-1} \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1}) + \mathbf{Q}_{12} \mathbf{Q}_{22.1}^{-1} \\ &= -\mathbf{Q}_{11} \mathbf{Q}_{11.2}^{-1} \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} + \mathbf{Q}_{12} \mathbf{Q}_{22.1}^{-1}. \end{aligned}$$

Using the Schur complement identity $\mathbf{Q}_{11} \mathbf{Q}_{11.2}^{-1} = \mathbf{I} + \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} \mathbf{Q}_{21} \mathbf{Q}_{11.2}^{-1}$ and further simplification via the Woodbury identity confirms this equals $\mathbf{0}$.

11.2 Why this matters for regression

Multiplying the block inverse by $\mathbf{Q}_{XY}$:

$$\begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \mathbf{Q}_{XX}^{-1} \mathbf{Q}_{XY} = \begin{pmatrix} \mathbf{Q}_{11.2}^{-1} & -\mathbf{Q}_{11.2}^{-1} \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} \\ -\mathbf{Q}_{22.1}^{-1} \mathbf{Q}_{21} \mathbf{Q}_{11}^{-1} & \mathbf{Q}_{22.1}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{Q}_{1Y} \\ \mathbf{Q}_{2Y} \end{pmatrix}.$$

For β_1 :

$$\begin{aligned} \beta_1 &= \mathbf{Q}_{11.2}^{-1} \mathbf{Q}_{1Y} - \mathbf{Q}_{11.2}^{-1} \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} \mathbf{Q}_{2Y} \\ &= \mathbf{Q}_{11.2}^{-1} (\mathbf{Q}_{1Y} - \mathbf{Q}_{12} \mathbf{Q}_{22}^{-1} \mathbf{Q}_{2Y}) \\ &= \mathbf{Q}_{11.2}^{-1} \mathbf{Q}_{1Y.2}, \end{aligned}$$

confirming equation (13).

12 Summary of Key Formulas

Quantity	Bivariate	Multivariate
BLP slope	$\frac{\text{Cov}(X, Y)}{\text{Var}(X)}$	$\mathbb{E}[XX']^{-1}\mathbb{E}[XY]$
BLP intercept	$\mathbb{E}[Y] - \beta\mathbb{E}[X]$	(absorbed into X when $1 \in X$)
Orthogonality	$\mathbb{E}[Xe] = 0$	$\mathbb{E}[Xe] = \mathbf{0}_{k \times 1}$
Error variance	$\text{Var}(Y) - \beta^2 \text{Var}(X)$	$Q_{YY} - \mathbf{Q}_{YX} \mathbf{Q}_{XX}^{-1} \mathbf{Q}_{XY}$
Sub-vector coeff.	—	$\mathbf{Q}_{11 \cdot 2}^{-1} \mathbf{Q}_{1Y \cdot 2}$
OVb	$\frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)} \beta_2$	$\Gamma_{12} \beta_2$

In each case, the multivariate formula reduces to the bivariate formula when $k = 1$ (plus a constant). The matrix algebra is the same operation—projecting and residualizing—applied to vectors instead of scalars.